

SaNaML Practice Datasets: Codebook and Exercises

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About these datasets

All five files are simulated. No real person, customer, student or employee appears in them. Effects were planted on purpose, so you can check whether your analysis finds what was built in. The files are CSV, so they open in SaNaML (Data screen, Upload), Excel and any other software. Missing values are blank cells.

File	Use it for	Rows
practice_students.csv	Statistics: tests, ANOVA, regression, reliability, data quality, machine learning	400
practice_marketing.csv	Marketing analytics: A/B test, regression on spend, churn prediction, segmentation	1,500
practice_brand_survey.csv	Marketing survey: reliability, scale scores, group comparison	500
practice_hr.csv	HR analytics: pay equity, attrition prediction, engagement drivers, training evaluation	1,200
practice_engagement.csv	HR survey: reliability, scale scores, department comparison	600

Also shipped with SaNaML: segments_demo.csv (clustering), lv_scores_demo.csv (machine learning on latent variable scores), utaut_items_demo.csv and careless_demo.csv (data quality).

Tip: the ID column (StudentID, CustomerID, RespondentID, EmployeeID) is a label. Do not use it as a variable.

1. practice_students.csv (a university study)

Variable	Meaning
StudentID	Identifier
Gender, Age, Year	Female/male, 17 to 24, year of study 1 to 4
Programme	Business, Science, Arts

Variable	Meaning
Residence	Home, Hostel, Rented (3% missing)
Teaching_method	Lecture, Blended, Flipped (students were assigned at random)
Study_hours_week	Hours studied per week (five extreme values on purpose)
Sleep_hours	Hours per night (about 3 to 4% missing)
Anxiety_score	0 to 60, right-skewed (about 3 to 4% missing)
Pre_score, Post_score	Test before and after the course (paired)
Score_T1, Score_T2, Score_T3	Same students tested three times (repeated measures)
Exam_score	0 to 100 (about 3 to 4% missing)
Project_score	Second outcome for MANOVA
Passed	Yes if Exam_score is 50 or more
Pass_before, Pass_after	Pass status on the pre and post test (paired categories)
Satisfaction_1to5	Ordinal rating (about 3 to 4% missing)
Mot1 to Mot6	Motivation items, 1 to 5
Anx1 to Anx6	Anxiety items, 1 to 5. Anx3 and Anx5 are reverse-keyed on purpose, so recode them (6 minus value) before scoring

Eight students answered every item with 3 (straight-lining). The data quality screen should flag them.

What is planned, and where to look:

- Descriptive statistics and Charts: Anxiety_score is right-skewed; Study_hours_week has outliers (box plot).
- Independent-samples t-test and Mann-Whitney: Exam_score by Gender. Women score about 4 points higher.
- Paired t-test and Wilcoxon: Pre_score against Post_score (large gain, about 6 points).
- One-way ANOVA, Welch, Tukey, Kruskal-Wallis: gain (Post minus Pre) differs by Teaching_method (Flipped largest).
- Repeated-measures ANOVA and Friedman: Score_T1, T2, T3 (steady rise).
- Factorial ANOVA, ANCOVA, MANOVA: Exam_score and Project_score by Teaching_method and Gender; Pre_score as covariate.
- Chi-square and Fisher: Gender by Programme (more women in Arts); Passed by Teaching_method.
- McNemar: Pass_before against Pass_after.
- Correlation and regression: Exam_score on Study_hours_week, Sleep_hours, Anxiety_score and Gender (R-squared about .17). Hierarchical: add Gender in step 2.

- Logistic regression: Passed on the same predictors.
- Reliability and scale scores: Mot1 to Mot6 (alpha about .86) and Anx1 to Anx6, with Anx3 and Anx5 reverse-coded. The items were built from two underlying factors, Motivation and Anxiety, which correlate negatively (SaNaML has no factor-analysis screen; use the construct scores).
- Machine learning: predict Passed or Exam_score, compare models, compare with the regression.
- Clustering: cluster students on Study_hours_week, Sleep_hours, Anxiety_score.
- Power analysis: plan a new study from the effect sizes you found.

2. practice_marketing.csv (a retailer's customers)

Variable	Meaning
CustomerID	Identifier
Age, Gender, Region	Customer profile
Income_k	Household income in thousands (about 5% missing, skewed)
Tenure_months	Months as a customer
Loyalty_member	Yes/No
Main_channel	Store, Online, Mobile app
Visits_per_month, Avg_basket, Discount_share	Shopping behaviour
Complaints_12m	Count of complaints
Satisfaction_1to10, Recommend_0to10	Survey ratings (about 5% missing)
Annual_spend	Spend in the year (skewed)
Campaign_group	Standard or New offer (random assignment)
Responded	Yes/No, responded to the campaign
Monthly_spend_before, Monthly_spend_after	Spend around the campaign (paired)
Churned	Yes/No, left within the year (about 17%)

What is planted:

- A/B test: Responded by Campaign_group (chi-square). New offer about 44%, Standard about 26%.
- Paired t-test: Monthly_spend_before against after.
- Regression: Annual_spend on Income_k, Tenure_months, Loyalty_member, Satisfaction_1to10, Visits_per_month (R-squared about .4 for log(1 + spend), against about .8 for raw spend). Try a transformation because of the skew.
- Group comparison: Satisfaction by Main_channel (ANOVA), Annual_spend by Loyalty_member (Mann-Whitney).

- Logistic regression and machine learning: Churned from satisfaction, complaints, tenure and loyalty. Responded from offer, loyalty and channel. Compare models and check the checks screen.
- Clustering: segment customers on Visits_per_month, Avg_basket, Discount_share, Satisfaction_1to10, Tenure_months; describe clusters by Region, Channel and Churned.
- Correlation and heat map: spend, income, visits and basket move together.

3. practice_brand_survey.csv (brand perception survey)

Trust1 to Trust4, Value1 to Value4, Loyalty1 to Loyalty3, all rated 1 to 7, plus Age_group, Gender and Customer_type. Trust and Value are correlated; Loyalty depends on both. Value2 and Loyalty2 have 3% missing. Ten respondents straight-lined (answered every item with 5).

- Data quality: find the straight-liners and remove them.
- Reliability: each scale has alpha near .85 or higher.
- The items were built from three underlying factors (Trust, Value, Loyalty). Make a scale score for each.
- Compute scale scores, then compare Loyalty by Customer_type (ANOVA) and Age_group; regress Loyalty on Trust and Value.
- Clustering and machine learning on the scale scores.

4. practice_hr.csv (a company's employees)

Variable	Meaning
EmployeeID	Identifier
Department	Sales, Operations, Finance, IT, HR
Job_level	1 to 5
Gender, Age, Education	Profile (Diploma, Bachelor, Master, Doctorate)
Tenure_years	Years with the company
Annual_salary	Skewed. After controls, women earn about 4% less (planted gap)
Performance_rating	1 to 5
Training_hours, Overtime_hours_month, Commute_km, Remote_days_week	Work conditions
Manager_support, Work_life_balance, Engagement, Job_satisfaction	Survey scores, 1 to 5
Absence_days	Count
Promoted_last_3y	Yes/No (about 11%)
Skills_test_before, Skills_test_after	Training evaluation (paired)
Engagement_Q1 to Engagement_Q4	Engagement in four quarters (repeated)

Variable	Meaning
	measures)
Left_company	Yes/No, left within the year (about 15%)
• Pay equity: Annual_salary by Gender (t-test, Mann–Whitney); regression of log salary on level, tenure, department, education and gender. • Training evaluation: paired t-test on the skills test; correlate gain with Training_hours. • Engagement drivers: regression of Engagement on Manager_support, Work_life_balance, Overtime and Performance. • Department differences: ANOVA and Kruskal–Wallis on Engagement or Absence_days; chi-square of Left_company by Department. • Attrition: logistic regression and machine learning on Left_company; engagement, overtime, tenure and promotion matter. The outcome is imbalanced (about 15%), so look at sensitivity and precision, not only accuracy. • Repeated measures: Engagement_Q1 to Q4. • Clustering: employee types on engagement, overtime, tenure and salary. • Missing data: Manager_support, Commute_km and Job_satisfaction have about 3 to 4% missing.	

When you report results about groups of people, keep small groups private and read results as patterns, not as decisions about individuals.

5. practice_engagement.csv (engagement survey)

Vigor1 to Vigor3, Dedication1 to Dedication3, Absorption1 to Absorption3 (1 to 7), plus Department, Gender, Tenure_band and Works_remotely.

- Reliability for the three scales and for all nine items together. The items were built from three correlated factors.
- Compare scale scores by Department (HR highest, then Finance and IT, with Sales and Operations lowest), by Gender and by Works_remotely.
- Add the scale scores to the data and run clustering.

Suggested order for learners

1. Load practice_students.csv, run the data quality screen, then describe the variables and draw histograms and box plots.
2. Work through the tests in the order of the Analyses menu, using the planted effects above as the answer key.
3. Repeat the same ideas on the marketing and HR files, then try machine learning and clustering.
4. Add each result to the Word report.

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